HH-I-551E
March 11, 1974
SUPERSEDING
Fed. Spec. HH-I-551D
January 13, 1969

FEDERAL SPECIFICATION

INSULATION BLOCK AND BOARDS, THERMAL (CELLULAR GLASS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers foamed-cellular glass for use as thermal insulation and for special purposes.

1.2 Classification.

1.2.1 Types. Cellular glass insulation shall be furnished in the following types, as specified (see 6.2):

Type I - Flat blocks.
Type II - Pipe covering (see 6.5).
Type III - Special shapes (see 6.5).
Type IV - Roof boards.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposals, unless otherwise specified, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-B-790 - Building Paper Vegetable Fiber: (Kraft, Waterproofed, Water Repellent and Fire Resistant).
PPP-B-601 - Box, Wood, Cleated-plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-corner.
PPP-B-636 - Box, Fiberboard.

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, and Standards, at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards, from established distribution point in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

American Society for Testing and Materials (ASTM) Standards:

C165 - Compressive Strength of Preformed Block-Type Thermal Insulation.
C168 - Standard Definitions of Terms Relating to Thermal Insulating Materials.
C177 - Thermal Conductivity of Materials by Means of the Guarded Hot Plate.
C203 - Breaking Load and Calculated Flexural Strength of Preformed Block-Type Thermal Insulation.
C240 - Cellular Glass Insulation Block.
C303 - Density of Preformed Block-Type Thermal Insulation.
C355 - Water Vapor Transmission of Thick Materials.
C518 - Thermal Conductivity of Materials by Means of the Heat Flow Meter.
C550 - Trueness and Squareness of Block Thermal Insulation.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

National Motor Freight Traffic Association, Inc. Agent

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The material shall consist of a glass composition which has been foamed or cellulated under molten conditions, annealed and set to form a rigid material with hermetically sealed cells.

3.2 Physical properties. Physical properties of type I and type IV insulation shall conform to the requirements of table I.

TABLE I. Physical properties

Density, LBS/ft^3(kg/m^3) 1/	
Minimum	7.0(112)
Maximum	9.5(152)
Compressive Strength, PSI(N/m^2) 2/	
Minimum	75(5.7x10^5)
Flexural Strength(modulus of rupture)	
Minimum PSI(N/m^2)	60(4.15x10^5)
Water Absorption	
Percent by Volume (average)	0.5
Water Vapor Permeability, at 80°F.(26.7°C)	
Perm-inch (metric perm-cm) 3/	
Maximum	0.005(0.0084)
Thermal Conductivity (k), at 75°F.(23.9°C)	
B.t.u.-in./ħ.ft.^{2}·°F.(W/m°C) 4/	0.415(0.60)

1/ Lbs/ft.^{3}(kg/m^3)- Pounds per cubic foot (kilogram per cubic meter).
2/ PSI(N/m^2)- Pounds per square inch (Newtons per square meter).
3/ Perm-inch (metric per-cm) - grains per hour per square foot per inch of Hg. per inch of thickness (grains per 24 hrs. per square meter per millimeter of Hg. per centimeter of thickness).
4/ B.t.u.-in./ft.^{2}.°F.(W/m°C) - British thermal units per inch of thickness per hour per square foot per degree Fahrenheit (Watts per meter per degree centrigrade).

3.3 Dimensional tolerances. Dimensional tolerances for length, width, and thickness of material furnished under this specification shall conform to table II.

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TABLE II. Dimensional tolerances in inches (millimeters)

Dimension	Block		Roof Board	
	Inches	Millimeter	Inches	Millimeter
Length	±0.625	1.59	±0.125	3.18
Width	±0.625	1.59	±0.625	1.59
Thickness	±0.625	1.59	±0.625	1.59

3.3.1 Squareness of blocks and roof boards. Blocks and roof boards shall not be out of square more than 0.0625 inch per foot (1.59 mm per 0.305 meter).

3.4 Dimensions of material. All materials furnished under this specification shall conform to the manufacturers standard available sizes for each type.

3.4.1 Type I blocks. Blocks shall be rectangular sections and shall be true to form and dimension, the corners square and the sides and ends parallel. Size and thickness of blocks shall conform to table III as specified (see 6.2). Dimensions shall be determined in accordance with ASTM C-303.

TABLE III. Size and thickness in inches (millimeters) of block insulation

Width		Length		Thickness									
in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
12	305	18	457	1.5	38.1	1.75	44.5	2	50.8	2.5	63.5	4	102	5	127
18	457	24	610	-	-	-	-	2	50.8	2.5	63.5	4	102	5	127

3.4.2 Type IV roof boards. Roof boards shall be 24 inches (610 mm) wide by 48 inches (1219mm) long by 1.5 inches (38.1mm), 1.75 inches (44.45mm), or 2 inches (50.8mm) thick as specified (see 6.2). The conductance (C value) of roof boards at 75° F. (23.9°C) mean temperature, shall be no greater than 0.28 B.t.u./h·ft.^{2}·°F. (1.59 W/m^{2}·°C) for 1.5 inch (38.1mm), 0.24 B.t.u./h·ft.^{2}·°F (1.36 W/m^{2}·°C) for 1.75 inch (44.5mm), and 0.21 B.t.u./h·ft.^{2}·°F (1.19 W/m^{2}·°C) for 2 inch (50.8mm). Roof boards shall be faced on both sides with waterproofed building paper conforming to UU-B-790, type I, class C. Building paper shall be applied to surface of the roof board with hot asphalt.

3.5 Standard commercial product. The insulation block and boards shall as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the insulation being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest product model.

3.6 Workmanship. The insulation block and boards shall have no visible defects such as broken corners, edges, separations, blow holes greater than a half-inch, or other harmful defects that will adversely affect its service qualities.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection lot. Unless otherwise specified (see 6.2), a lot shall consist of all insulation of the same type, and size offered to the Government at one time for purposes of inspection.

4.2.2 Sampling for visual and dimensional examination. A random sample of insulation shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-3. The acceptable quality level (AQL) shall be 2.5 percent defective for major defects.

4.2.3 Sampling for tests. A representative sample of insulation shall be selected from each lot offered in accordance with MIL-STD-105 at inspection level S-1 and shall be based on an AQL of 6.5 percent defective. Block samples shall be of sufficient size to perform the tests specified in 4.4.

4 3 Visual and dimensional examination. Samples selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table IV.

TABLE IV. Classification of defects

Categories	Examination	Requirements paragraphs
Major:		
101	Type not as specified.	1.2 and 6.5
102	Length, width, and thickness not with tolerance.	3.3, 4.3.1 3.4.1, 3.4.2
103	Edges not square by more than 0.0625 inch per foot.	3.2.1, 3.4.1
104	Broken corners, or edges.	3.4.2
105	Separation or blow holes greater than one-half inch.	3.6

4.3.1 Length, width, thickness and squareness of blocks and roof boards. Sample blocks and roof boards selected in accordance with 4.2.2 shall have the length, width, and thickness dimensions measured in accordance with ASTM C303. The squareness of the blocks and boards shall be determined in accordance with ASTM C550.

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4.4 Performance tests. Samples selected in accordance with 4.2.3 shall be tested as specified herein. Any sample failing to pass any test shall be considered a defective unit in reference to that specific test.

4.4.1 Density. The density of the insulation shall be determined in accordance with ASTM C303, except that no drying will be necessary.

4.4.2 Compressive (crushing) strength. Compressive strength shall be determined in accordance with ASTM C165 with the samples prepared in accordance with ASTM C240.

4.4.3 Flexural strength (modulus of rupture). Flexural strength shall be determined in accordance with ASTM C240 and ASTM C203.

4.4.4 Thermal conductivity. Three specimens shall be tested to determine the thermal conductivity value. The maximum average conductivity value for the three specimens shall not be greater than the value specified in Table I when tested as specified in ASTM C177 or ASTM C518. If the inspection of the first sampling shows failure to conform to the requirements of this specification, a second set of three specimens from the same lot shall be tested and the results of this retest averaged with the results of the original test. Upon retest, failure to conform to this specification shall constitute grounds for rejection.

4.4.5 Water absorption. Water absorption shall be determined in accordance with ASTM C240.

4.4.6 Water vapor permeability. Water vapor permeability shall be determined in accordance with ASTM C355.

4.5 Certificate of compliance. Unless otherwise specified (see 6.2), a certificate of compliance may be accepted as proof that the requirements of 4.4 through 4.4.6 of this specification have been met.

4.6 Inspection of preparation for delivery. The packing and marking of insulation shall be insepcted to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The insulation shall be packed in close fitting wood cleated plywood or nailed wooden boxes conforming to PPP-B-601 (overseas type) and PPP-B-621 (class 2), respectively. Each piece of insulation shall be protected from abrasion and from contact with each other by the use of slipsheets made of paper having a minimum basis weight of 60 pounds (27.21kg.).

5.1.2 Level B. The insulation shall be packed in close fitting fiberboard boxes conforming to PPP-B-636, type CF, domestic class, variety SW, grade 275. Each piece of insulation shall be protected from abrasion and from contact with each other by the use of slipsheets made of paper having a minimum basis weight of 60 pounds (27.21kg.).

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5.1.2.1 When specified (see 6.2), the box shall conform to PPP-B-636, V3c.

5.1.3 Level C. The insulation shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or National Motor Freight Classification rules.

5.2 Marking (see 6.2).

5.2.1 Civil agencies. Shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Cellular-glass insulation covered by this specification is intended for building insulation, including roofs, walls, core walls, floors, ceilings, on-grade slabs, perimeters, and refrigerated spaces.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type of insulation required (see 1.2 and 6.5).

c. Block size (width and length) and thickness required (see 3.4.1 and Table III).

d. Required dimensions for roof boards (see 3.4.2 and 6.3).

e. Lot size, if different than specified (see 4.2.1).

f. If certificate of compliance is not acceptable (see 4.5).

g. Level of Packing required (see 5.1).

h. If V3c fiberboard is required for level B packing (see 5.1.2.1).

i. Marking required (see 5.2).

6.3 Roofing boards. Roofing which is made up of roof boards adhered to a steel deck with not more than 25 pounds per 100 square feet of hot asphalt is rated class 1 construction by Factory Mutual Engineering Division of the Associated Factory Mutual Fire Insurance Companies.

6.4 Fire hazard information. Type I and type IV are inert materials, intrinsically noncombustible.

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6.5 Classification change. The classification and cross-reference from HH-I-551D to HH-I-551E are as follows:

HH-I-551D	HH-I-551E	HH-I-1751/3a
Type I	Type I	
Type II Class 1 Class 2 Class 3 Class 4		Type I Class 1 Class 2 Class 3 Class 4
Type III		Type II
Type IV	Type IV	

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activity:

Army - CE

User activities:

Navy - SH, CG

Preparing activity:
Navy - YD

CIVIL AGENCY COORDINATING ACTIVITIES:

COMMERCE - NBS
GSA - PCD
GSA - FSS
HUD - TCS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. ents each.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

HH-I-551E: Insulation Block and Boards, Thermal (Cellular Glass)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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